

Due monday, November 25th

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1. Unitary matrices and operators.

(a) [2 pts.] Look up and report the definition of a unitary matrix.

(b) [1 pt.] Show that the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is its own inverse.

Hint: no need to calculate an inverse explicitly; use the fact/definition that a matrix multiplied by its inverse gives the identity.

(c) [5 pts.] Show that the three Pauli matrices are unitary.

(Comment: We have seen previously that they are also hermitian. These matrices are both hermitian and unitary!)

(d) [3 pts.] Show that the two-dimensional vector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ has the same norm as $\sigma_y \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$, i.e., operating with σ_y keeps the norm unchanged.

(Note: This is a special case of the fact that a unitary operation keeps the norm unchanged.)

(e) [4 pts.] If $\hat{U}$ is a unitary operator, show that the state $|\psi\rangle = \hat{U}|\phi\rangle$ has the same norm as the state $|\phi\rangle$.

2. The state $|\mu_1\rangle$ is an eigenstate of $\hat{M}$ with corresponding eigenvalue m_1 .

(a) [2 pts.] Show that $|\mu_1\rangle$ is also an eigenstate of $\hat{M}^2$, and find the corresponding eigenvalue. Show that $|\mu_1\rangle$ is also an eigenstate of $\hat{M}^n$, where n is an integer, and find the corresponding eigenvalue.

(b) [5 pts.] How is the operator $e^{\hat{M}}$ defined?

How is $e^{\alpha\hat{M}}$ defined, if α is a complex number?

Is $|\mu_1\rangle$ is an eigenstate of $e^{\alpha\hat{M}}$? If so, find the corresponding eigenvalue.

3. Review the properties of the Dirac delta function.

We will consider a particle in one dimension subject to a negative (attractive) delta potential at $x = 0$:

$$V(x) = -\lambda\delta(x)$$

This system has a single bound state at some negative energy, $E < 0$. We will find the energy and wavefunction of this bound state.

- (a) [**5 pts.**] Consider the left half-line ($x < 0$). Write down the general solution of the Schroedinger equation in this region. Remember: we are considering $E < 0$.

Argue why one of the terms of the general solution can be dropped.

- (b) [**2 pts.**] Similarly find the general solution on the right half-line ($x > 0$), and identify the term that should be dropped.

- (c) [**3 pts.**] Explain why the derivative of the wavefunction, $\psi'(x)$, need not be continuous at $x = 0$.

- (d) [**6 pts.**] Let ϵ be an infinitesimally small positive number. By integrating both sides of the Schroedinger equation from $-\epsilon$ to ϵ , show that the derivative of the wavefunction has the following discontinuity around the $x = 0$ point:

$$\psi'(\epsilon) - \psi'(-\epsilon) = -\frac{2m\lambda}{\hbar^2}\psi(0) + \text{infinitesimal term}$$

- (e) [**8 pts.**] Determine the energy of the bound state and the constants in the wavefunction.

You can use continuity of $\psi(x)$ and the normalization of the full wavefunction. In addition, the relation above for the discontinuity of $\psi'(x)$ can be used as a boundary condition:

$$\psi'_R(0) - \psi'_L(0) = -\frac{2m\lambda}{\hbar^2}\psi_L(0)$$

where $\psi_L(x)$ and $\psi_R(x)$ are the wavefunctions on the left half-line and right half-line respectively.

- (f) [**4 pts.**] Plot the wavefunction as a function of position. Make sure the (dis)continuity at $x = 0$ is clearly visible.